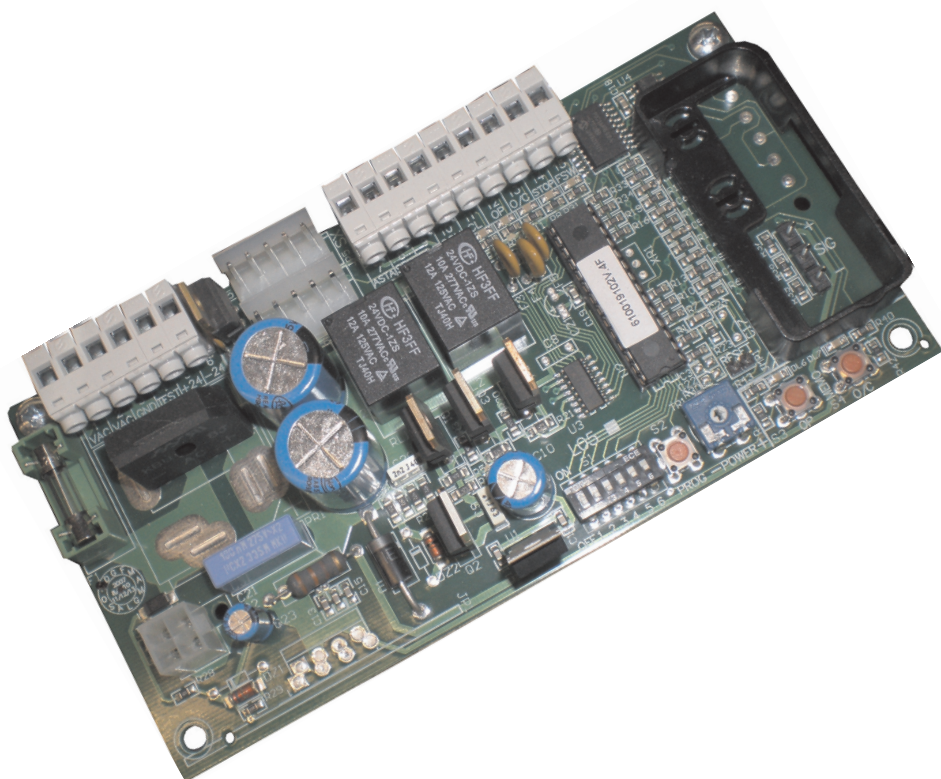


E604



FAAC

ITALIANO

AVVERTENZE PER L'INSTALLATORE OBBLIGHI GENERALI PER LA SICUREZZA



ATTENZIONE! È importante per la sicurezza delle persone seguire attentamente tutta l'istruzione. Una errata installazione o un errato uso del prodotto può portare a gravi danni alle persone.

1. Leggere attentamente le istruzioni prima di iniziare l'installazione del prodotto.
2. I materiali dell'imballaggio (plastica, polistirolo, ecc.) non devono essere lasciati alla portata dei bambini in quanto potenziali fonti di pericolo.
3. Conservare le istruzioni per i riferimenti futuri.
4. Questo prodotto è stato progettato e costruito esclusivamente per l'utilizzo indicato in questa documentazione. Qualsiasi altro utilizzo non espressamente indicato potrebbe pregiudicare l'integrità del prodotto e/o rappresentare fonte di pericolo.
5. FAAC declina qualsiasi responsabilità derivata dall'uso improprio o diverso da quello per cui l'automatismo è destinato.
6. Non installare l'apparecchio in atmosfera esplosiva: la presenza di gas o fumi infiammabili costituisce un grave pericolo per la sicurezza.
7. Gli elementi costruttivi meccanici devono essere in accordo con quanto stabilito dalle Norme EN 12604 e EN 12605.
8. Per i Paesi extra-CEE, oltre ai riferimenti normativi nazionali, per ottenere un livello di sicurezza adeguato, devono essere seguite le Norme sopra riportate.
9. FAAC non è responsabile dell'insorgenza della Buona Tecnica nella costruzione delle chiusure da motorizzare, nonché delle deformazioni che dovessero intervenire nell'utilizzo.
10. L'installazione deve essere effettuata nell'osservanza delle Norme EN 12453 e EN 12445. Il livello di sicurezza dell'automazione deve essere C+D.
11. Prima di effettuare qualsiasi intervento sull'impianto, togliere l'alimentazione elettrica e scollegare le batterie.
12. Prevedere sulla rete di alimentazione dell'automazione un interruttore onnipolare con distanza d'apertura dei contatti uguale o superiore a 3 mm. È consigliabile l'uso di un magnetotermico da 6A con interruzione onnipolare.
13. Verificare che a monte dell'impianto vi sia un interruttore differenziale con soglia da 0,03 A.
14. Verificare che l'impianto di terra sia realizzato a regola d'arte e collegare le parti metalliche della chiusura.
15. L'automazione dispone di una sicurezza intrinseca antischacciamento costituita da un controllo di coppia. È comunque necessario verificare la soglia di intervento secondo quanto previsto dalle Norme indicate al punto 10.
16. I dispositivi di sicurezza (norma EN 12978) permettono di proteggere eventuali aree di pericolo da Rischi meccanici di movimento, come ad Es. schiacciamento, convogliamento, cesaioamento.
17. Per ogni impianto è consigliato l'utilizzo di almeno una segnalazione luminosa nonché di un cartello di segnalazione fissato adeguatamente sulla struttura dell'impianto, oltre ai dispositivi citati al punto 16.
18. FAAC declina ogni responsabilità ai fini della sicurezza e del buon funzionamento dell'automazione, in caso vengano utilizzati componenti dell'impianto non di produzione FAAC.
19. Per la manutenzione utilizzare esclusivamente parti originali FAAC.
20. Non eseguire alcuna modifica sui componenti facenti parte del sistema d'automazione.
21. L'installatore deve fornire tutte le informazioni relative al funzionamento manuale del sistema in caso di emergenza e consegnare all'Utente l'installazione dell'impianto il libretto d'avvertenze allegato al prodotto.
22. Non permettere ai bambini o persone di sostare nelle vicinanze del prodotto durante il funzionamento.
23. L'applicazione non può essere utilizzata da bambini, da persone con ridotte capacità fisiche, mentali, sensoriali o da persone prive di esperienza o del necessario addestramento.
24. Tenere fuori dalla portata dei bambini radiocomandi o qualsiasi altro datore di impulso, per evitare che l'automazione possa essere azionata involontariamente.
25. Il transito tra le ante deve avvenire solo a cancello completamente aperto.
26. L'utente utilizzatore deve astenersi da qualsiasi tentativo di riparazione o d'intervento e deve rivolgersi solo ed esclusivamente a personale qualificato FAAC o centri d'assistenza FAAC.
27. Tutto quello che non è previsto espressamente in queste istruzioni non è permesso.

ENGLISH

IMPORTANT NOTICE FOR THE INSTALLER GENERAL SAFETY REGULATIONS



ATTENTION! To ensure the safety of people, it is important that you read all the following instructions. Incorrect installation or incorrect use of the product could cause serious harm to people.

1. Carefully read the instructions before beginning to install the product.
2. Do not leave packing materials (plastic, polystyrene, etc.) within reach of children as such materials are potential sources of danger.
3. Store these instructions for future reference.
4. This product was designed and built strictly for the use indicated in this documentation. Any other use, not expressly indicated here, could compromise the good condition/operation of the product and/or be a source of danger.
5. FAAC declines all liability caused by improper use or use other than that for which the automated system was intended.
6. Do not install the equipment in an explosive atmosphere: the presence of inflammable gas or fumes is a serious danger to safety.
7. The mechanical parts must conform to the provisions of Standards EN 12604 and EN 12605.
8. For non-EU countries, to obtain an adequate level of safety, the Standards mentioned above must be observed, in addition to national legal regulations.
9. FAAC is not responsible for failure to observe Good Technique in the construction of the closing elements to be motorised, or for any deformation that may occur during use.
10. The installation must conform to Standards EN 12453 and EN 12445. The safety level of the automated system must be C+D.
11. Before attempting any job on the system, cut out electrical power and disconnect the batteries.
12. The mains power supply of the automated system must be fitted with an all-pole switch with contact opening distance of 3mm or greater. Use of a 6A thermal breaker with all-pole circuit break is recommended.
13. Make sure that a differential switch with threshold of 0.03 A is fitted upstream of the system.
14. Make sure that the earthing system is perfectly constructed, and connect metal parts of the means of the closure to it.
15. The automated system is supplied with an intrinsic anti-crushing safety device consisting of a torque control. Nevertheless, its tripping threshold must be checked as specified in the Standards indicated at point 10.
16. The safety devices (EN 12978 standard) protect any danger areas against mechanical movement Risks, such as crushing, dragging, and shearing.
17. Use of at least one indicator-light is recommended for every system, as well as a warning sign adequately secured to the frame structure, in addition to the devices mentioned at

point "16".

18. FAAC declines all liability as concerns safety and efficient operation of the automated system, if system components not produced by FAAC are used.
19. For maintenance, strictly use original parts by FAAC.
20. Do not in any way modify the components of the automated system.
21. The installer shall supply all information concerning manual operation of the system in case of an emergency, and shall hand over to the user the warnings handbook supplied with the product.
22. Do not allow children or adults to stay near the product while it is operating.
23. The application cannot be used by children, by people with reduced physical, mental, sensorial capacity, or by people without experience or the necessary training.
24. Keep remote controls or other pulse generators away from children, to prevent the automated system from being activated involuntarily.
25. Transit through the leaves is allowed only when the gate is fully open.
26. The User must not in any way attempt to repair or to take direct action and must solely contact qualified FAAC personnel or FAAC service centres.
27. Anything not expressly specified in these instructions is not permitted.

FRANÇAIS

CONSIGNES POUR L'INSTALLATEUR RÈGLES DE SÉCURITÉ



ATTENTION! Il est important, pour la sécurité des personnes, de suivre à la lettre toutes les instructions. Une installation erronée ou un usage erroné du produit peut entraîner de graves conséquences pour les personnes.

1. Lire attentivement les instructions avant d'installer le produit.
2. Les matériaux d'emballage (matière plastique, polystyrène, etc.) ne doivent pas être laissés à la portée des enfants car ils constituent des sources potentielles de danger.
3. Conserver les instructions pour les références futures.
4. Ce produit a été conçu et construit exclusivement pour l'usage indiqué dans cette documentation. Toute autre utilisation non expressément indiquée pourrait compromettre l'intégrité du produit et/ou représenter une source de danger.
5. FAAC décline toute responsabilité qui dériverait d'usage impropre ou différent de celui auquel l'automatisme est destiné.
6. Ne pas installer l'appareil dans une atmosphère explosive: la présence de gaz ou de fumées inflammables constitue un grave danger pour la sécurité.
7. Les composants mécaniques doivent répondre aux prescriptions des Normes EN 12604 et EN 12605.
8. Pour les Pays extra-CEE, l'obtention d'un niveau de sécurité approprié exige non seulement le respect des normes nationales, mais également le respect des Normes susmentionnées.
9. FAAC n'est pas responsable du non-respect de la Bonne Technique dans la construction des fermetures à motoriser, ni des déformations qui pourraient intervenir lors de l'utilisation.
10. L'installation doit être effectuée conformément aux Normes EN 12453 et EN 12445. Le niveau de sécurité de l'automatisme doit être C+D.
11. Couper l'alimentation électrique et déconnecter la batterie avant toute intervention sur l'installation.
12. Prévoir, sur le secteur d'alimentation de l'automatisme, un interrupteur onnipolaire avec une distance d'ouverture des contacts égale ou supérieure à 3 mm. On recommande d'utiliser un magnétothermique de 6A avec interruption onnipolaire.
13. Vérifier qu'il y ait, en amont de l'installation, un interrupteur différentiel avec un seuil de 0,03 A.
14. Vérifier que la mise à terre est réalisée selon les règles de l'art et y connecter les pièces métalliques de la fermeture.
15. L'automatisme dispose d'une sécurité intrinsèque anti-écrasement, formée d'un contrôle du couple. Il est toutefois nécessaire d'en vérifier le seuil d'intervention suivant les prescriptions des Normes indiquées au point 10.
16. Les dispositifs de sécurité (norme EN 12978) permettent de protéger des zones éventuellement dangereuses contre les Risques mécaniques du mouvement, comme l'écrasement, l'acheminement, le cisaillement.
17. On recommande que toute installation soit dotée au moins d'une signalisation lumineuse, d'un panneau de signalisation fixe, de manière appropriée, sur la structure de la fermeture, ainsi que des dispositifs cités au point "16".
18. FAAC décline toute responsabilité quant à la sécurité et au bon fonctionnement de l'automatisme si les composants utilisés dans l'installation n'appartiennent pas à la production FAAC.
19. Utiliser exclusivement, pour l'entretien, des pièces FAAC originales.
20. Ne jamais modifier les composants faisant partie du système d'automatisme.
21. L'installateur doit fournir toutes les informations relatives au fonctionnement manuel du système en cas d'urgence et remettre à l'Usager qui utilise l'installation les "Instructions pour l'Usager" fournies avec le produit.
22. Interdire aux enfants ou aux tiers de stationner près du produit durant le fonctionnement.
23. Ne pas permettre aux enfants, aux personnes ayant des capacités physiques, mentales et sensorielles limitées ou dépourvues de l'expérience ou de la formation nécessaires d'utiliser l'application en question.
24. Eloigner de la portée des enfants les radiocommandes ou tout autre générateur d'impulsions, pour éviter tout actionnement involontaire de l'automatisme.
25. Le transit entre les vantaux ne doit avoir lieu lorsque le portail est complètement ouvert.
26. L'utilisateur doit s'abstenir de toute tentative de réparation ou d'intervention et doit s'adresser uniquement et exclusivement au personnel qualifié FAAC ou aux centres d'assistance FAAC.
27. Tout ce qui n'est pas prévu expressément dans ces instructions est interdit.

ESPAÑOL

ADVERTENCIAS PARA EL INSTALADOR REGLAS GENERALES PARA LA SEGURIDAD



ATENCIÓN! Es sumamente importante para la seguridad de las personas seguir atentamente las presentes instrucciones. Una instalación incorrecta o un uso impropio del producto puede causar graves daños a las personas.

1. Leer detenidamente las instrucciones antes de instalar el producto.
2. Los materiales del embalaje (plástico, poliestireno, etc.) no deben dejarse al alcance de los niños, ya que constituyen fuentes potenciales de peligro.
3. Guardar las instrucciones para futuras consultas.
4. Este producto ha sido proyectado y fabricado exclusivamente para la utilización indicada en el presente manual. Cualquier uso diverso del previsto podría perjudicar el funcionamiento del producto y/o representar fuente de peligro.
5. FAAC declina cualquier responsabilidad derivada de un uso impropio o diverso del previsto.
6. No instalar el aparato en atmósfera explosiva: la presencia de gas o humos inflamables constituye un grave peligro para la seguridad.
7. Los elementos constructivos mecánicos deben estar de acuerdo con lo establecido en las Normas EN 12604 y EN 12605.
8. Para los países no pertenecientes a la CEE, además de las referencias normativas

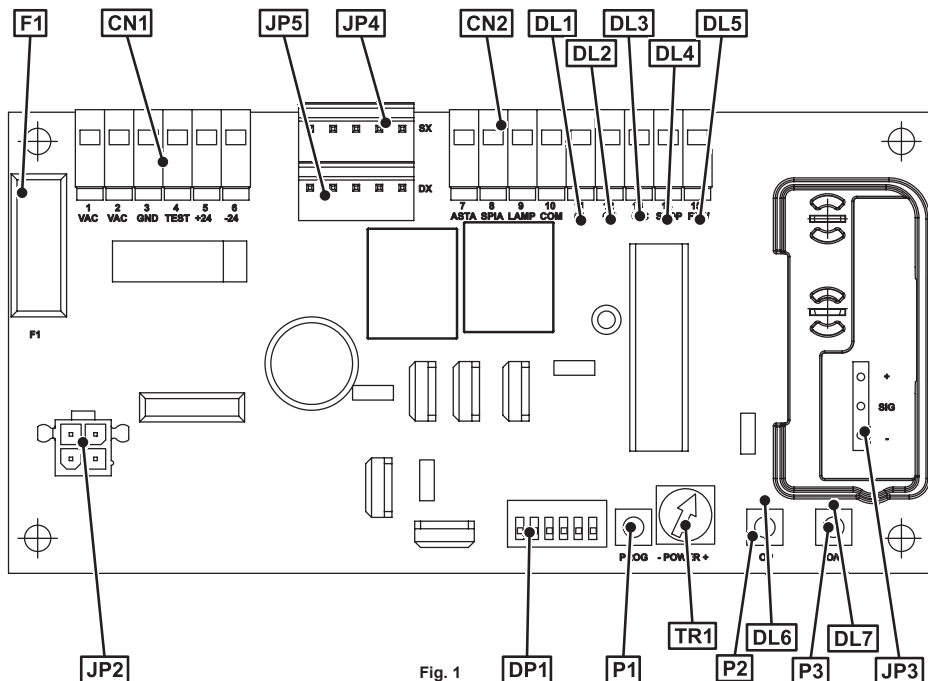


Fig. 1

Trasformatore toroidale - Toroidal transformer
 Transformateur toroïdal - Transformador toroïda
 toroid-transformator - Ringtransformator

230 V~ 50Hz
 115 V~ 60Hz

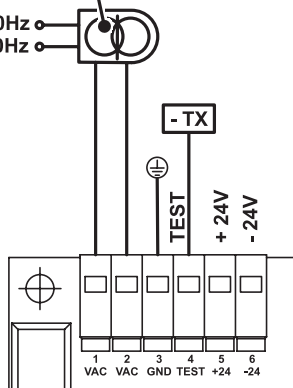


Fig. 2

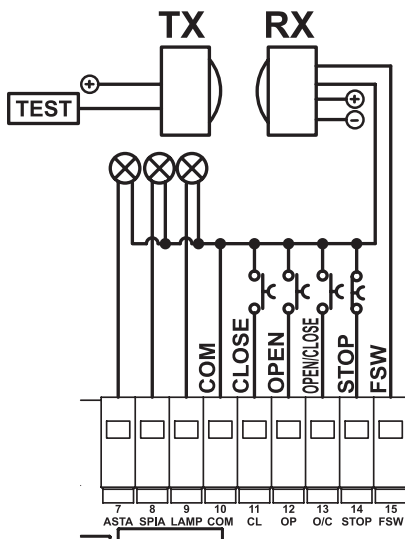


Fig. 3





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CE DECLARATION OF CONFORMITY

Manufacturer: FAAC S.p.A.

Address: Via Calari, 10 - 40069 - Zola Predosa - Bologna - ITALY

Declares that: Control unit mod. **E604 24V**

- conforms to the essential safety requirements of the following EEC directives:
 - 2006/95/EC Low Voltage directive.
 - 2004/108/EC Electromagnetic Compatibility directive.

Additional information:

This product underwent a test in a typical uniform configuration (all products manufactured by FAAC S.p.A.).

Bologna, December 30, 2009

Managing Director
A. Marcellan

Notes on reading the instruction

Read this installation manual to the full before you begin installing the product.

The symbol  indicates notes that are important for the safety of persons and for the good condition of the automated system.

The symbol  draws your attention to the notes on the characteristics and operation of the product.



1. DESCRIPTION

Thank you for choosing our product. FAAC is sure that it will give you all the performances you are looking for. All our products are the result of a long experience in the field of the automated systems.

Thanks to its high powered microprocessor, this 24 V= control unit for automatic barriers offers a wide range of adjustments, including deceleration and motor control.

Thanks to its integrated encoder, the control unit constantly controls rod position and movement, intervening as soon as a faulty situation is detected.

The **FAILSAFE** function makes it possible to detect any faults in the operation of the photocells.

The main operating modes and functions are set by a dip-switch, whereas the adjustment of work and pause times is done by self learning during programming.

2. TECHNICAL SPECIFICATIONS

Power supply voltage	230 V~ (+6% -10%) 50 Hz / 115 V~ 60 Hz
Supply voltage of control unit	24 V~ nominal ①
Absorbed power	3 W
Motor nominal power	48 W
Accessories max. load	500 mA
Flashing lamp max. load	15 W
Max. load for beam lights	15 W
Max. load for courtesy light/indicator light	5 W
Operating ambient temperature	↓-20°C ↑+55°C
Protective fuse	F1=8 A F2=500 mA F3=630 mA (resettable)
Function logics	Automatic - Condo - Manual
Opening / closing time	Through self-learning during programming
Pause time	Through self-learning during programming
Obstacle detection	Trimmer-adjustable
Selectable functions	Type of logic - Operation of open command - No mains power supplied function - Beam lights flashing - Safety devices behaviour
Terminal board inputs	Power supply - Earth - Close - Open - Open/Close - Stop - Safety devices - FAILSAFE
Inputs with connector	Battery kit - Radio connector
Terminal board outputs	Power supply for accessories - On beam lights - Indicator light - Flash-light
Board dimensions	79 x 158 mm
Characteristics of optional batteries	See kit on price list



① According to mains voltage, there may be different output values on the board power feed terminals. Before putting into service, always check if the output voltage on the secondary winding of the transformer is from 20 V~ to 26 V~. Voltage must always be measured load free.

3. PREPARATIONS



To ensure people's safety, all warnings and instructions in this booklet must be carefully observed. Incorrect installation or incorrect use of the product could cause serious harm to people.

- Make sure that an adequate differential switch is installed upstream of the system as specified by current regulations.
- On the main power supply, install a thermal breaker with omnipolar switching.
- Make sure that an adequate earthing system is available.
- To lay cables, use adequate rigid and/or flexible tubes.
- Always separate the connecting cables of low voltage accessories from those supplying 230/115 V~, using separate sheaths to avoid possible interference.

4. BOARD LAY-OUT (Fig. 1)

COMPONENTS		COMPONENTS	
CN1	Power supply terminal-board	P3	OPEN/CLOSE channel memory storage push-button
CN2	Commands terminal-board	F1	Board protection fuse
JP2	Battery kit connector	DL1	CLOSE contact LED
JP3	Connector for receiver module	DL2	OPEN contact LED
JP4	Power supply to RH closing motor	DL3	OPEN/CLOSE contact LED
JP5	Power supply to LH closing motor	DL4	STOP contact LED
TR1	Trimmer for adjusting obstacle detection sensitivity	DL5	Photocells status LED
P1	Programming push-button	DL6	OPEN contact signalling LED
P2	OPEN channel memory storage push-button	DL7	OPEN/CLOSE contact signalling LED



5. DESCRIPTION OF CONNECTIONS

5.1. TERMINAL BOARD CN1 (Fig. 2)

5.1.1. POWER SUPPLY

Terminals "1 & 2". The secondary winding wires of the toroidal transformer must be connected to these terminals.

 **Before making this connection, check if the voltage at the transformer output is in the range from 20 V~ to 26 V~**

5.1.2. EARTHING

Terminal "3". To this terminal, connect the earthing wire coming from the terminal inside the motor body.

 **The connection is absolutely necessary for correct operation of the control unit.**

5.1.3. FAILSAFE

Terminal "4". Connect the negative terminal of the photocell transmitters to this terminal. This contact is used by the Failsafe function to check the correct operation of the photocells before every closing manoeuvre of the beam. To activate this function refer to paragraph 6.

5.1.4. POWER SUPPLY FOR ACCESSORIES

Terminals "5 & 6". 24 V~ output max. 500 mA for feeding the external accessories.

 **Maximum load of this output is 500 mA.**

 **Observe the power supply polarity.**

5.2. TERMINAL BOARD CN2 (Fig. 3)

5.2.1. BEAM LIGHTS

Terminals "7 & 10". Output 24 V~ max. 15 W. Connect the power cables of the luminous strip to these terminals. The luminous strip lights on when the beam is open or closed, whereas it flashes when the beam is moving.

 **Terminal "10" is the negative polarity output.**

5.2.2. INDICATOR LIGHT

Terminals "8 & 10". Output 24 V~ max. 5 W. An indicator light, if any, should be connected to these terminals. The indicator light makes it possible to see the beam status from a remote position, e.g. a porter's lodge - details:

- Beam opening is identified by fast flashing.
- When the beam is open, the indicator light stays lighted on a steady light-beam.
- Beam closing is identified by slow flashing.
- When the beam is closed, the indicator light stays OFF.

 **24 V~ max. 5 W lamps can be connected to this output.**

 **Terminal "10" is the negative polarity output.**

5.2.3. FLASHING LIGHT

Terminals "9 & 10". Output 24 V~ max. 15 W. A steady beam flashing light is connected to these terminals - flashing is controlled by the control unit. The flashing light is active while the beam is moving, whereas when the beam is idle, whether open or closed, it stays OFF. Pre-flashing of 0.5 sec was inserted before the beam opening manoeuvre, to signal that the beam is about to move. In addition to signalling beam movement, the flashing light signals - by a series of flashes - that the beam is being powered by the buffer battery (optional).

 **Terminal "10" is the negative polarity output.**

5.2.4. CLOSE

Terminals "10 & 11". Normally open contact. Connect, between these 2 terminals, any pulse generator (e.g. push-button, key selector, etc...) which, by closing the contact, commands beam closure only. The status of this input is signalled by LED "DL1".

 **The CLOSE command is not active during programming.**

 **If there are several pulse generators, they must be connected in parallel.**

5.2.5. OPEN

Terminals "10 & 12". Normally open contact. Connect, between these 2 terminals, any pulse generator (e.g. push-button, key selector, etc...) which, by closing the contact, commands beam opening only. The status of this input is signalled by LED "DL2".

 **The OPEN command is not active during programming.**

 **If there are several pulse generators, they must be connected in parallel.**

5.2.6. OPEN / CLOSE

Terminals "10 & 13". Normally open contact. Connect, between these 2 terminals, any pulse generator (e.g. push-button, key selector, etc...) which, by closing the contact, commands beam opening and/or closing. The behaviour of this input is defined by dip-switch 2, see paragraph 6. The status of this input is signalled by LED "DL3".

 **If there are several pulse generators, they must be connected in parallel.**



5.2.7. STOP

Terminals "10 & 14". Normally closed contact. Connect, between these 2 terminals, any pulse generator (e.g. push-button, key selector, etc..) which, by opening the contact, must command immediate stop of the beam and the de-activation of any automatic re-closure. To resume the normal programmed cycle after this contact has been activated, use any pulse generator which commands the opening and/or closing of the beam. The status of this input is signalled by LED "DL4".

 If there are several pulse generators, they must be connected in series.

5.2.8. SAFETY DEVICES

Terminals "10 & 15". Normally closed contact. Connect, to these terminals, any safety device (e.g. photocells) which, by opening the contact, acts on beam motion. They can be active only during closure, or active in both closing and opening, according to the position of dip-switch 6, see paragraph 6.

 If the safety devices are not used, a connection must be made between terminal 4 and terminal 15.

 If there are several safety devices, they must be connected in series.

Safety devices active at closure

If the safety devices are activated during closure, the control unit immediately reverses beam movement until opening is complete without disabling - if selected - automatic re-closing of the beam.

Safety devices active at closure and opening:

In this case the safety devices are active on both beam movements. If the safety devices are activated during closure, the control unit immediately reverses beam movement until opening is complete without disabling - if selected - automatic re-closing of the beam. If the safety devices are activated during opening, the control unit immediately stops beam movement and keeps it stopped until the safety device is reset (obstacle removed), and only then will the opening manoeuvre which had been started be resumed.

The status of this input is signalled by the LED located under the contact.

5.3 MOTOR CONNECTION

The automated system is supplied designed for **right hand** closure, with the motor connected to connector **JP4**. For **left hand** closure, the motor must be connected to connector **JP5**.

To identify beam closing direction, look at the automated system from the side of the release device. If the beam has to descend on the left of the automated system in order to close, left hand closure is involved, vice-versa, if the beam has to descend on the right of the automated system in order to close, right hand closure is involved.

 Two motors cannot be connected to the same control unit.

5.4. BATTERY KIT

A battery kit can be connected to the board - see price list - to make up for a possible power cut.

For correct positioning of the battery kit inside the motor body, refer to the instruction for the mechanical part.

The battery kit must be connected to connector **JP2**.

6. DIP-SWITCH ADJUSTMENT

6 dip-switches are located on the control unit. They command the behaviour of the automated system and of the safety devices connected to it. The behaviour of every dip-switch is summarised on the following table:

Automatic closing: Allows you to switch any automatic closing of the bar



Automatic closure enabled



Automatic closing deactivated

Admission OPEN / CLOSE: Allows you to select the input behavior OPEN / CLOSE



Open / Stop / Close / Stop / Open / ...



Open / Close / Open / ...

Condominial Logic: Allows you to activate the pool. Enabling this feature during the opening operation of the bar, the control unit ignores any commands OPEN or OPEN / CLOSE



Logic condominium activated



Logic condominium deactivated

Rod Open: This feature is described in the following paragraph



Function open rod activated



Function open auction deactivated



Phototest: Activating this function, the unit carries out before closing the bar, a test of Switch operation of the photocells. This feature is available in power from version 4 onwards.

ON OFF		Phototest activated
ON OFF		Phototest disabled
Photocells: With this dip-switch you can select the mode of action of the photocells		
ON OFF		Photocell is active during closing and opening
ON OFF		Photoelectric cells are active in closing

6.1. OPEN BEAM FUNCTION

With this function, combined or not with the battery kit, the following beam behaviour patterns are possible:

WITHOUT BATTERY KIT

- Dip-switch 4=OFF:** If there is a mains power cut, the beam stays stopped in the position it is in. When mains power returns, after 2 seconds the control unit automatically commands the beam to close, preparing itself for normal operation.
- Dip-switch 4=ON:** If there is a mains power cut, the beam stays stopped in the position it is in. When mains power returns, a pulse must be sent to ensure that the control unit resumes normal operation.

WITH BATTERY KIT

- Dip-switch 4=OFF:** If there is a mains power cut, the beam continues operating normally. At every opening, the flashing light flashes twice consecutively, at intervals of 3 seconds between the flashes, for a maximum time of 30 seconds, signalling that the beam is battery powered. When mains power is resumed, the automated system resumes normal operation.

The number of cycles possible with the automated system battery powered, depends on the battery charge level, on the time elapsing since mains power was stopped, and on the conditions of the automated system, external temperature etc...

- Dip-switch 4=ON:** In the event of a mains power cut, the automated system automatically opens the beam, stopping it in vertical position and disabling all the commands. When mains power returns, if automatic logic was selected, the automated system automatically performs a closing movement preparing itself for normal operation. Vice-versa, if manual logic was selected, the automated system waits for a pulse to resume normal operation.

7. MEMORY STORING THE RADIO CODE

The control board has an integrated 2-channel decoding system (**DS, SLH, LC/RC**) named **OMNIDEC**. This system makes it possible to memory-store both **OPEN** command and the **OPEN/CLOSE** command of the automated system - this is made possible by an additional receiver module and radio controls on the same frequency.

The 3 types of radio codes (DS, LSH, LC/RC) cannot coexist. Only one radio code can be used at a time. To change over from one code to another, you must delete the existing one (see paragraph 7.4), and repeat the memory-storage procedure.

Fitting and, if necessary, removing the receiver module must be done only after cutting power to the board.

The receiver module can only be inserted in one position. Orient the module correctly without forcing.

7.1. Memory storage of DS radio controls

A maximum of **two codes** can be stored. One on the **OPEN** channel and one on the **OPEN/CLOSE** channel.

- On the **DS** radio control, select the required ON-OFF combination for the 12 dip-switches.
- On the control unit, press the push-button of the channel you wish to memory store, **P2** for the **OPEN** channel or **P3** for the **OPEN/CLOSE** channel.
- The relevant LED on the control unit begins to flash - release the push-button.
- On the radio control, press the push-button with which you wish to associate the selected channel.
- The LED on the control unit lights up on steady beam for about one second, signalling that the radio control was stored in the memory, then it resumes flashing.
- To add other radio controls or memory store the second channel, repeat the operations from point 1

7.2. Memory storage of SLH radio controls

You can memory store up to a maximum of **250 codes**, subdivided between the two channels, **OPEN** and **OPEN/CLOSE**.

- On the radio control, simultaneously press and hold down push-buttons **P1** and **P2** (see radio control instructions).
- After about one second, the LED of the radio control begins to flash.
- Release both push-buttons.
- Press and hold down push-button **P2** or **P3** on the board to respectfully memory store the **OPEN** or **OPEN/CLOSE** channel. The relevant LED begins to flash.
- Simultaneously press the push-button of the radio control with which you wish to associate the selected command.
- Check if the LED relating to the command being memory stored (**DL6** for the **OPEN** channel or **DL7** for the **OPEN/CLOSE** channel) lights up on steady beam for about two seconds to confirm correct memory storage.
- To finish programming, press twice in close succession, the push-button of the memory stored radio control.

The automated system will perform an opening manoeuvre - make sure that there are no obstacles inside the operating range.



8. To memory store the other channel, repeat all the procedure from point 1.



To add other radio controls, transfer the code of the memory-stored push-button of the radio control to the relevant push-button of the radio controls to be added, observing the following procedure:

- On the memory stored radio control, simultaneously press and hold down push-buttons **P1** and **P2** (see radio control instructions).
- The radio control LED begins to flash.
- Release both push-buttons.
- Put the two radio controls frontally **into contact**.
- On the memory stored radio control, press and hold down the push-button relating to the channel you wish to transfer - the radio control LED lights up on steady beam.
- On the radio control to be memory stored, press the required push-button and release it after the radio control has flashed twice.
- To finish programming, press twice in close succession, the push-button of the memory stored radio control.

 **The automated system will perform an opening manoeuvre - make sure that there are no obstacles inside the operating range.**

7.3. Memory storage of LC/RC radio controls



You can memory store up to a maximum of **250 codes**, subdivided between the two channels, **OPEN** and **OPEN/CLOSE**.

1. Use **LC/RC** remote controls only with receiver module at **433 MHz**.
2. On the control unit, press the push-button of the channel you wish to memory store, **P2** for the **OPEN** channel or **P3** for the **OPEN/CLOSE** channel.
3. The relevant LED on the control unit begins to flash - release the push-button.
4. On the radio control, press the push-button with which you wish to associate the selected channel.
5. The LED on the control unit lights up on steady beam for about one second, signalling that the radio control was stored in the memory, then it resumes flashing.
6. During this stage further radio controls can be stored in the memory.
7. After about 10 seconds, the control unit automatically exits the learning stage.
8. To add other radio controls or memory store the second channel, repeat the operations from point 1

7.3.1. REMOTE MEMORY STORAGE OF LC/RC RADIO CONTROLS

Other radio controls can be remotely stored only with the **LC/RC** radio controls, i.e. without using the push-buttons of the control unit., but using a previously stored radio control.

1. Get a radio control already memory stored on one of the 2 channels.
2. Step near to the automated system.
3. Press and hold down push-buttons **P1** and **P2** (see radio control instructions) simultaneously for about 5 seconds.
4. Within 5 seconds, press, on the memory stored radio control, the push-button you wish to transfer to the new radio control. In this way the learning stage on the selected channel is activated on the control unit.
5. Within 5 seconds, press, on the new radio control, the push-button you wish to associate with the selected channel.
6. After the new radio control has been stored in the memory, the control unit keeps the learning mode active on the selected channel for about 5 seconds.
7. During these 5 seconds, other radio controls can be memory stored on the control unit, as ever associated with the activated channel.
8. When 5 seconds have elapsed from memory-storage of the last radio control, the control unit automatically exits the learning stage.
9. To check if the radio control was correctly memory stored, wait for 5 seconds after sending the code.

7.4. Deletion of radio codes

To delete **all** the radio controls stored in the memory, go through the following procedure:

1. Press and hold down one of the two push-buttons **P2** or **P3**.
2. The relevant LED begins to flash.
3. After five seconds, the LED starts to flash at high speed.
4. After another five seconds both LEDs, **DL6** and **DL7** light up on steady beam.
5. Release the push-button.

 **This operation is irreversible, and all radio controls associated with both the OPEN and the OPEN/CLOSE command will be deleted.**

8. OBSTACLE DETECTION SENSITIVITY

The obstacle detection device enables the control unit to detect any obstacles while the beam is moving. As this is an electronic device, it stays constant through time and is not subject to variations - this guarantees an automated system with a constant safety level.

By using the **TR1** trimmer, you can increase or reduce the obstacle detection sensitivity. Turn the trimmer **clockwise** to **increase** detection sensitivity, vice-versa turn it **anti-clockwise** to **reduce** obstacle detection sensitivity.

This device is active both during the beam closing stage and during the beam opening stage.

If it operates during the opening stage, it stops the operation of the automated system and reverses motion for 1 second. At this point, a pulse must be sent to the control unit, to resume normal programmed operation.

If it operates during the closing stage, it reverses operation until the automated system is completely open without disabling any automatic closing.

If it operates three times in succession, it blocks the operation of the automated system and locates itself in stop position, disabling automatic closing, if it was activated. To restore normal operation, you must command the beam to open or close. Behaviour patterns will differ, according to the command sent:

- **OPEN** or **OPEN/CLOSE** pulse: In this case, the beam starts an opening manoeuvre at slower speed until the beam is completely open. When complete opening has been reached, the beam resumes its normal operating cycle, re-activating automatic re-closing, if it had been enabled.



- **CLOSE** pulse: In this case, the beam starts a closing manoeuvre at slower speed until the beam is completely closed. When closed, the beam resumes its normal operating cycle, re-activating automatic re-closing, if it had been enabled.

9. PROGRAMMING

 **When the control unit is first powered up, a programming cycle must be performed. The board keeps all commands disabled, with the exception of programming push-button P1, and of the pulse generators connected to the OPEN/CLOSE input**

When you have made the connections of all the accessories and generators, program the operating cycle. To program the control unit, follow the instructions below:

1. Turn the differential switch of the control unit, to make sure that the automated system is not powered.
2. Release the automated system, using the release device - see instructions for the mechanical part - and position the beam at about half of its opening angle (45°).
3. Re-lock the beam and make sure that it cannot be moved manually.
4. Power up the system.
5. Press **P1** for about 1 second, the flashing lamp goes on at steady beam
6. Supply an **OPEN/CLOSE** command by means of any device connected to this input: the beam starts moving.

 **The first manoeuvre the beam carries out is in closing. If the automated system starts an opening manoeuvre, check if the motor connector is correctly connected, as described in paragraph 5.3.**

7. Restore power and repeat the procedure from point 5.
8. When the closing mechanical stop is reached, the beam begins the opening manoeuvre.
9. When maximum opening is achieved, the pause time count for automatic re-closure begins.
10. When the required time has elapsed, supply another **OPEN/CLOSE** pulse, and the automated system will begin the closing stage.
11. When the closing position is reached, the flashing lamp goes off and programming has finished.

 **During the programming procedure, the beam moves more slowly and the safety devices are disabled.**

10. CONTROL LEDs

There are 7 control LEDs on the control unit. The meanings of the LEDs are shown on the table below:

INPUT LEDs		
LED	ON	OFF
DL1 - CLOSE Input	Active command	Inactive command
DL2 - OPEN input	Active command	Inactive command
DL3 - OPEN/CLOSE input	Active command	Inactive command
DL4 - STOP input	Inactive command	Active command
DL5 - FSW input	Safety devices free	Safety devices engaged
PROGRAMMING LEDs		
LED	ON	OFF
DL6 - OPEN channel	Active input	Inactive input
DL7 - OPEN/CLOSE channel	Active input	Inactive input

 **The bold print indicates the condition of the LEDs with the control unit powered up and the beam closed in rest position.**

**11. FUNCTION LOGICS****LOGIC A (Automatic) DIP-SWITCH 1=ON / DIP-SWITCH 3=OFF**

BARRIER STATUS	INPUTS						
	OPEN	CLOSE	OPEN / CLOSE		STOP	PHOTOCELLS	
			DIP-SWITCH 2=ON	DIP-SWITCH 2=OFF		DIP-SWITCH 6=OFF	DIP-SWITCH 6=ON
CLOSED	Opens and re-closes after pause time	No effect	Opens and re-closes after pause time	Opens and re-closes after pause time	No effect (if active, it disables all commands)	No effect	Disables OPEN commands
OPEN IN PAUSE	No effect	Closes immediately	Stops operation and, at next pulse, closes	Re-closes immediately	Stops operation	Disables commands at release resets pause time	Disables commands at release resets pause time
AT OPENING	No effect	Closes	Stops operation and, at next pulse, closes	Reverses motion on closing	Stops operation	No effect	Stops movement and restarts after release
AT CLOSING	Reverses on opening	No effect	Stops operation and, at next pulse, opens	Reverses motion on opening	Stops operation	Reverses motion on opening	Stops movement and opens at release

LOGIC M (Manual) DIP-SWITCH 1=OFF / DIP-SWITCH 3=OFF

BARRIER STATUS	INPUTS						
	OPEN	CLOSE	OPEN / CLOSE		STOP	PHOTOCELLS	
			DIP-SWITCH 2=ON	DIP-SWITCH 2=OFF		DIP-SWITCH 6=OFF	DIP-SWITCH 6=ON
CLOSED	Opens	No effect	Opens	Opens	No effect (if active, it disables all commands)	No effect	Disables OPEN commands
OPEN	No effect	Closes immediately	Re-closes immediately	Re-closes immediately	Stops operation	Disables all commands	Disables commands
AT OPENING	No effect	Closes	Stops operation and, at next pulse, closes	Reverses motion on closing	Stops operation	No effect	Stops movement and restarts after release
AT CLOSING	Reverses on opening	No effect	Stops operation and, at next pulse, opens	Reverses motion on opening	Stops operation	Reverses motion on opening	Stops movement and opens at release


LOGIC C (Condo) DIP-SWITCH 1=ON / DIP-SWITCH 3=ON

BARRIER STATUS	INPUTS						
	OPEN	CLOSE	OPEN / CLOSE		STOP	PHOTOCELLS	
			DIP-SWITCH 2=ON	DIP-SWITCH 2=OFF		DIP-SWITCH 6=OFF	DIP-SWITCH 6=ON
CLOSED	Opens and re-closes after pause time	No effect	Opens and re-closes after pause time	Opens and re-closes after pause time	No effect (if active, it disables all commands)	No effect	Disables OPEN commands
OPEN IN PAUSE	No effect	Closes immediately	Reloads pause time	Re-closes immediately	Stops operation	Disables commands and at release resets pause time	Disables commands and at release resets pause time
AT OPENING	No effect	Reverses motion on closing	No effect	No effect	Stops operation	No effect	Stops movement and restarts after release
AT CLOSING	Reverses on opening	No effect	Reverses motion on opening	Reverses motion on opening	Stops operation	Reverses motion on opening	Stops movement and opens at release

12. DISPOSAL


The trash container symbol shown on the product label, on the product or in the instructions indicates that the product, at the end of its life, must be disposed of separately from other waste products.

The user must therefore bring the equipment that has reached the end of its life to an authorised electrical and electronic waste disposal centre, or return it to the dealer when new equivalent equipment is purchased, on a one-to-one basis.

Proper separate collection for the recycling, treatment or compatible environmental disposal of the equipment no longer in use contributes to avoiding negative effects on the environment and health and favours reuse and/or recycling of the materials with which the equipment is made.

Unlawful disposal of the product by the user is subject to fines provided for by the regulations in force.

nacionales, para obtener un nivel de seguridad adecuado, deben seguirse las Normas arriba indicadas.

9. FAAC no es responsable del incumplimiento de las buenas técnicas de fabricación de los cierres que se han de motorizar, así como de las deformaciones que pudieran intervenir en la utilización.
10. La instalación debe ser realizada de conformidad con las Normas EN 12453 y EN 12445. El nivel de seguridad de la automatización debe ser C+D.
11. Quitar la alimentación eléctrica y desconectar las baterías antes de efectuar cualquier intervención en la instalación.
12. Colocar en la red de alimentación de la automatización un interruptor omnipolar con distancia de apertura de los contactos igual o superior a 3 mm. Se aconseja usar un magnetotérmico de 6A con interruptor omnipolar.
13. Comprobar que la instalación disponga línea arriba de un interruptor diferencial con umbral de 0,03 A.
14. Verificar que la instalación de tierra esté correctamente realizada y conectar las partes metálicas del cierre.
15. La automatización dispone de un dispositivo de seguridad antiaplastamiento constituido por un control de par. No obstante, es necesario comprobar el umbral de intervención según lo previsto en las Normas indicadas en el punto 10.
16. Los dispositivos de seguridad (norma EN 12978) permiten proteger posibles áreas de peligro de Riesgos mecánicos de movimiento, como por ej. aplastamiento, arrastre, corte.
17. Para cada equipo se aconseja usar por lo menos una señalización luminosa así como un cartel de señalización adecuadamente fijado a la estructura del bastidor, además de los dispositivos indicados en el "16".
18. FAAC declina toda responsabilidad relativa a la seguridad y al buen funcionamiento de la automatización si se utilizan componentes de la instalación que no sean de producción FAAC.
19. Para el mantenimiento utilicen exclusivamente piezas originales FAAC.
20. No efectúen ninguna modificación en los componentes que forman parte del sistema de automatización.
21. El instalador debe proporcionar todas las informaciones relativas al funcionamiento del sistema en caso de emergencia y entregar al usuario del equipo el manual de advertencias que se adjunta al producto.
22. No permitan que niños o personas se detengan en proximidad del producto durante su funcionamiento.
23. La aplicación no puede ser utilizada por niños, personas con reducida capacidad física, mental, sensorial o personas sin experiencia o la necesaria formación.
24. Mantengan lejos del alcance los niños los telemandos o cualquier otro emisor de impulso, para evitar que la automatización pueda ser accionada involuntariamente.
25. Sólo puede transitar entre las hojas si la cancela está completamente abierta.
26. El usuario debe abstenerse de intentar reparar o de intervenir directamente, y debe dirigirse exclusivamente a personal cualificado FAAC o a centros de asistencia FAAC.
27. Todo lo que no esté previsto expresamente en las presentes instrucciones debe entenderse como no permitido

22. Weder Kinder noch Erwachsene sollten sich während des Betriebs in der unmittelbaren Nähe der Automation aufhalten.
23. Die Anwendung darf nicht von Kindern, von Personen mit verminderter körperlicher, geistiger, sensorieller Fähigkeit oder Personen ohne Erfahrungen oder der erforderlichen Ausbildung verwendet werden.
24. Die Funktionseuerungen und alle anderen Impulsgeber sollten außerhalb der Reichweite von Kindern aufbewahrt werden, um ein versehentliches Aktivieren der Automation zu vermeiden.
25. Der Durchgang oder die Durchfahrt zwischen den Flügeln darf lediglich bei vollständig geöffnetem Tor erfolgen.
26. Der Benutzer darf direkt keine Versuche für Reparaturen oder Arbeiten vornehmen und hat sich ausschließlich an qualifiziertes Fachpersonal FAAC oder an Kundendienstzentren FAAC zu wenden.
27. Alle Vorgehensweisen, die nicht ausdrücklich in der vorliegenden Anleitung vorgesehen sind, sind nicht zulässig

NEDERLANDS

WAARSCHUWINGEN VOOR DE INSTALLATEUR ALGEMENE VEILIGHEIDSVOORSCHRIFTEN



LET OP! Het is belangrijk voor de veiligheid dat deze hele instructie zorgvuldig wordt opgevolgd. Een onjuiste installatie of foutief gebruik van het product kunnen ernstig persoonlijk letsel veroorzaken.

1. Lees de instructies aandachtig door alvorens te beginnen met de installatie van het product.
2. De verpakkingsmaterialen (plastic, polystyreen, enz.) mogen niet binnen het bereik van kinderen worden gelaten, want zij vormen een mogelijke bron van gevaar.
3. Bewaar de instructies voor raadpleging in de toekomst.
4. Dit product is uitsluitend ontworpen en gebouwd voor het doel dat in deze documentatie wordt aangegeven. Elk ander gebruik, dat niet uitdrukkelijk wordt vermeld, zou het product kunnen beschadigen en/of een bron van gevaar kunnen vormen.
5. FAAC aanvaardt geen enkele aansprakelijkheid voor schade die ontstaat uit oneigenlijk gebruik of ander gebruik dan waarvoor het automatische systeem is bedoeld.
6. Installeer het apparaat niet in een explosiegevaarlijke omgeving: de aanwezigheid van ontvlambare gasen of dampen vormt een ernstig gevaar voor de veiligheid.
7. De mechanische bouwelementen moeten in overeenstemming zijn met de bepalingen van de normen EN 12604 en EN 12605.
8. Voor niet-EEG landen moeten, om een goed veiligheidsniveau te bereiken, behalve de nationale voorschriften ook de bovenstaande normen in acht worden genomen.
9. FAAC is niet aansprakelijk als de regels der goede techniek niet in acht genomen zijn bij de bouw van het sluitwerk dat gemotoriseerd moet worden, noch voor vervormingen die zouden kunnen ontstaan bij het gebruik.
10. De installatie dient te geschieden in overeenstemming met de normen EN 12453 en EN 12445. Het veiligheidsniveau van het automatische systeem moet C+D zijn.
11. Alvorens ingrepen te gaan verrichten op de installatie moet de elektrische voeding worden weggenomen en moeten de batterijen worden afgekoppeld.
12. Zorg op het voedingsnet van het automatische systeem voor een meerpolige schakelaar met een opening tussen de contacten van 3 mm of meer. Het wordt geadviseerd een magnetothermische schakelaar van 6A te gebruiken met meerpolige onderbreking.
13. Controleer of er bovenstrooms van de installatie een differentieelschakelaar is geplaatst met een limiet van 0,03 A.
14. Controleer of de aardingsinstallatie vakdunk is aangelegd en sluit er de metalen delen van het sluitsysteem op aan.
15. Het automatische systeem beschikt over een intrinsieke beveiliging tegen inklemming, bestaande uit een controle van het koppel. De inschakelmoment hiervan dient echter te worden gecontroleerd volgens de bepalingen van de normen die worden vermeld onder punt 10.
16. De veiligheidsvoorzieningen (norm EN 12978) maken het mogelijk eventuele gevaarlijke gebieden te beschermen tegen Mechanische gevaren door beweging, zoals bijvoorbeeld inklemming, meesleuren of amputatie.
17. Het wordt voor elke installatie geadviseerd minstens één lichtsignaal te gebruiken alsook een waarschuwingsbord dat goed op de constructie van het hang- en sluitwerk dient te worden bevestigd, afgezien nog van de voorzieningen die genoemd zijn onder punt "16".
18. FAAC aanvaardt geen enkele aansprakelijkheid voor wat betreft de veiligheid en de goede werking van het automatische systeem, als er in de installatie gebruik gemaakt wordt van componenten die niet door FAAC zijn geproduceerd.
19. Gebruik voor het onderhoud uitsluitend originele FAAC-onderdelen.
20. Verriicht geen wijzigingen op componenten die deel uitmaken van het automatische systeem.
21. De installateur dient alle informatie te verstrekken over de handbediening van het systeem in noodgeval, en moet de gebruiker van de installatie het bij het product geleverde boekje met aanwijzingen overhandigen.
22. De toepassing mag niet worden gebruikt door kinderen, personen met lichamelijke, geestelijke en sensoriele beperkingen, of door personen zonder ervaring of de benodigde training.
23. Sta het niet toe dat kinderen of volwassenen zich ophouden in de buurt van het product terwijl dit in werking is.
24. Houd radio-afstandsbedieningen of alle andere impulsgevers buiten het bereik van kinderen, om te voorkomen dat het automatische systeem onopzettelijk kan worden aangegeven.
26. Ga alleen tussen de vleugels door als het hek helemaal geopend is.
26. De gebruiker mag zelf geen pogingen ondernemen tot reparaties of andere directe ingrepen, en dient zich uitsluitend te wenden tot gekwalificeerd en geautoriseerd FAAC-personeel of een erkend FAAC-servicecentrum.
27. Alles wat niet uitdrukkelijk in deze instructies wordt aangegeven, is niet toegestaan

DEUTSCH

HINWEIS FÜR DEN INSTALLATIONSTECHNIKER ALLGEMEINE SICHERHEITSVORSCHRIFTEN



ACHTUNG! Um die Sicherheit von Personen zu gewährleisten, sollte die Anleitung aufmerksam befolgt werden. Eine falsche Installation oder ein fehlerhafter Betrieb des Produktes können zu schwerwiegenden Personenschäden führen.

1. Bevor mit der Installation des Produktes begonnen wird, sollten die Anleitungen aufmerksam gelesen werden.
2. Das Verpackungsmaterial (Kunststoff, Styropor, usw.) sollte nicht in Reichweite von Kindern aufbewahrt werden, da es eine potentielle Gefahrenquelle darstellt.
3. Die Anleitung sollte aufbewahrt werden, um auch in Zukunft Bezug auf sie nehmen zu können.
4. Dieses Produkt wurde ausschließlich für den in diesen Unterlagen angegebenen Gebrauch entwickelt und hergestellt. Jeder andere Gebrauch, der nicht ausdrücklich angegeben ist, könnte die Unversehrtheit des Produktes beeinträchtigen und/oder eine Gefahrenquelle darstellen.
5. Die Firma FAAC lehnt jede Haftung für Schäden, die durch unsachgemäßen und nicht bestimmungsgemäßen Gebrauch der Automatik verursacht werden, ab.
6. Das Gerät sollte nicht in explosionsgefährdeten Umgebungen installiert werden: das Vorhandensein von entflammbaren Gasen oder Rauch stellt ein schwerwiegendes Sicherheitsrisiko dar.
7. Die mechanischen Bauelemente müssen den Anforderungen der Normen EN 12604 und EN 12605 entsprechen.
8. Für Länder, die nicht der Europäischen Union angehören, sind für die Gewährleistung eines entsprechenden Sicherheitsniveaus neben den nationalen gesetzlichen Bevorschriften die oben aufgeführten Normen zu beachten.
9. Die Firma FAAC übernimmt keine Haftung im Falle von nicht fachgerechten Ausführungen bei der Herstellung der anzutreibenden Schließvorrichtungen sowie bei Deformationen, die eventuell beim Betrieb entstehen.
10. Die Installation muß unter Beachtung der Normen EN 12453 und EN 12445 erfolgen. Die Sicherheitsstufe der Automatik sollte C+D sein.
11. Vor der Ausführung jeglicher Eingriffe auf der Anlage sind die elektrische Versorgung und die Batterie abzunehmen.
12. Auf dem Versorgungsnetz der Automatik ist ein omnipolarer Schalter mit Öffnungsabstand der Kontakte von über oder gleich 3 mm einzubauen. Darüber hinaus wird der Einsatz eines Magnetschutzschalters mit 6A mit omnipolarer Abschaltung empfohlen.
13. Es sollte überprüft werden, ob vor der Anlage ein Differentialschalter mit einer Auslöschschwelle von 0,03 A zwischengeschaltet ist.
14. Es sollte überprüft werden, ob die Erdungsanlage fachgerecht aufgeführt wurde. Die Metallteile der Schließung sollten an diese Anlage angeschlossen werden.
15. Die Automation verfügt über eine eingebaute Sicherheitsvorrichtung für den Quetschschutz, die aus einer Drehmomentkontrolle besteht. Es ist in jedem Falle erforderlich, deren Eingriffsschwelle gemäß der Vorgaben der unter Punkt "16" angegebenen Vorschriften zu überprüfen.
16. Die Sicherheitsvorrichtungen (Norm EN 12978) ermöglichen den Schutz eventueller Gefahrenbereiche vor mechanischen Bewegungsrisiken, wie zum Beispiel Quetschungen, Mitschleifen oder Schnittverletzungen.
17. Für jede Anlage wird der Einsatz von mindestens einem Lichtsignal empfohlen eines Hinweisschildes, das über eine entsprechende Befestigung mit dem Aufbau des Tors verbunden wird. Darüber hinaus sind die unter Punkt "16" erwähnten Vorrichtungen einzusetzen.
18. Die Firma FAAC lehnt jede Haftung hinsichtlich der Sicherheit und des störungsfreien Betriebs der Automatik ab, soweit Komponenten auf der Anlage eingesetzt werden, die nicht im Hause FAAC hergestellt wurden.
19. Bei der Instandhaltung sollten ausschließlich Originalteile der Firma FAAC verwendet werden.
20. Auf den Komponenten, die Teil des Automationssystems sind, sollten keine Veränderungen vorgenommen werden.
21. Der Installateur sollte alle Informationen hinsichtlich des manuellen Betriebs des Systems in Notfällen liefern und dem Betreiber der Anlage das Anleitungsbuch, das dem Produkt beigelegt ist, übergeben.

Le descrizioni e le illustrazioni del presente manuale non sono impegnative. La FAAC si riserva il diritto, lasciando inalterate le caratteristiche essenziali dell'apparecchiatura, di apportare in qualunque momento e senza impegnarsi ad aggiornare la presente pubblicazione, le modifiche che essa ritiene convenienti per miglioramenti tecnici o per qualsiasi altra esigenza di carattere costruttivo o commerciale.

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